$$\lambda = \frac{2\pi\hbar}{p}.$$

The implication is that to probe objects of ever smaller size one needs photons (or other objects) of ever larger momentum. Large momentum inevitably means large energy. It's for that reason that the discovery of the microscopic properties of matter required increasingly powerful particle accelerators.

8.3 FOURIER TRANSFORMS AND THE MOMENTUM BASIS

The wave function $\psi(x)$ has the important role of determining the probability for finding the particle at position x:

$$P(x) = \psi^*(x)\psi(x).$$

As we will see, no experiment can determine both the position and momentum of a particle simultaneously. But if we forego determining anything about the position, momentum can be measured precisely. The situation is quite analogous to that of the x and z components of a spin. Either value can be measured, but not both.

What is the probability that a particle has momentum p if we choose to measure it? The answer is a straightforward generalization of the principles laid down in Lecture 3. The probability that a momentum measurement will give momentum p is

$$P(p) = |\ \langle \mathbf{P}|\Psi\rangle\ |^2. \qquad (8.19)$$

The entity <**P**|Ψ> is called the wave function of |Ψ> in the *momentum representation*. Naturally, it is a function of p and is denoted by a new symbol: